

Exercise n. 1

The logistic equation is a classical nonlinear **Ordinary Differential Equation** (ODE) commonly used to model population growth in the presence of limited resources. The model captures the transition from an initial exponential growth phase to a saturation regime, where the solution approaches a finite carrying capacity. Let us consider the following Cauchy's problem involving a logistic equation:

$$\begin{cases} \frac{d\phi(t)}{dt} = r\phi(t)\left(1 - \frac{\phi(t)}{K}\right) \\ \phi(0) = \phi_0 = 0.5 \end{cases}$$

Where r is the intrinsic growth rate, while K is the carrying capacity. The analytical solution is given by:

$$\phi(t) = \frac{K}{1 + \frac{K - \phi_0}{\phi_0} e^{-rt}}$$

Write a numerical code to **solve the above nonlinear ODE and its initial condition over the time interval $t \in [0,10]$ with $r = 1$ and $K = 10$** . The code should be able to work both in **double and single precision**. The problem should be solvable using the following algorithms:

- I. **implicit Euler method;**
- II. **Crank-Nicolson method.**

The solution should be determined in the given time interval using both the aforementioned algorithms and with the following time steps, Δt :

1. $\Delta t = 1$;
2. $\Delta t = 10^{-1}$;
3. $\Delta t = 10^{-2}$;
4. $\Delta t = 10^{-3}$;
5. $\Delta t = 10^{-4}$.

Using the developed numerical code, perform the steps outlined below:

1. For the time step, $\Delta t = 1$, **plot the two numerical solutions as a function of time together with the analytical solution on a single graph.**
2. For both methods, **plot the absolute value of the error** (modulus of the difference between numerical and exact solution), evaluated at the last integration step, **as a function of the time step used for the numerical integration, $||e_N||$ vs Δt** . **Perform a fit of the obtained curves**

with a curve of the form $A_1 \cdot \Delta t^1$ for the implicit Euler method and $A_2 \cdot \Delta t^2$ for the Crank-Nicolson method. **Use a log-log scale.** Be careful to perform the fit avoiding constant and linear terms respectively, i.e., $A_1 \cdot \Delta t^1$ and **NOT** $A_1 \cdot \Delta t^1 + B_1$ and $A_2 \cdot \Delta t^2$ and **NOT** $A_2 \cdot \Delta t^2 + B_2 \Delta t + C_2$.

3. **Repeat the previous step in single precision** but avoid the fit.
4. **Answer the questions: what is the difference between double and single precision? What happens in the second case?**

TIPS: The problem is non-linear. Similarly to what was discussed in class, the numerical integration of the logistic equation requires, at each time step, the solution of a non-linear algebraic equation. For simplicity, this can be done using a Newton-Raphson (NR) algorithm, which can be easily implemented with the help of AI. Be careful to check, at each time step, the convergence of the NR algorithm and to allow for a sufficiently high number of sub-iterations (NR is an iterative method). Finally, recall that the NR method can be used to find the solution of an equation of the form

$g(\phi_{n+1}, \phi_n) = 0$ where the unknown is ϕ_{n+1} . The derivative $\frac{dg(\phi_{n+1}, \phi_n)}{d\phi_{n+1}}$ must be known. This

derivative can be derived analytically once the discrete form of the chosen time-integration scheme (implicit Euler or Crank-Nicolson) has been written.

OBS: The report can be prepared individually or in groups of up to three students. Conciseness, clarity of presentation, and the quality of figures are appreciated.